

Does earnings information predict abnormal returns?

A null result on 13,736 S&P 500 announcements, and the machinery built to make that answer trustworthy.

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ABSTRACT

Post-earnings-announcement drift is among the most-studied anomalies in finance and among the easiest to reproduce spuriously. This note tests whether information in earnings releases predicts short-horizon abnormal returns across 13,736 announcements by 466 S&P 500 companies, timestamped to the minute from SEC filings, with each year from 2019 to 2024 held out and predicted by a model frozen before it began. **It does not.** Mean out-of-sample rank IC is 0.022 ($t = 1.30$), the sector-neutral book returns a net Sharpe of -0.79, and alpha against Fama–French five factors plus momentum is -1.67% ($t = -2.16$). That Sharpe is uninformative rather than adverse: running the same book on the same events 200 times with the predictions shuffled yields -0.77 on average, placing the realised value at the 47th percentile of noise ($p = 0.53$). An earlier version of this study, on a quarter as many companies, reported a significant decay in skill; on the full sample the trend is +0.006 per year ($p = 0.61$) and does not replicate. Run on synthetic data with a known planted effect the same pipeline recovers it in 6/6 years at a net Sharpe of 4.82; with nothing planted it returns 0.022 and 0.37. That contrast is why the null above is worth believing rather than merely asserting.

1 · DESIGN

For each year Y , the model is fitted only on announcements whose 20-day outcome had fully resolved before 1 January Y , less a 25-session embargo; it is then frozen and used to predict every announcement in Y . The target is the cumulative abnormal return from **one session after** the announcement to twenty sessions after: windows beginning on the announcement day contain the opening gap, which cannot be traded, and booking it is the single largest source of overstated drift results. Events come from SEC 8-K Item 2.02 filings, so every one carries a minute-level acceptance timestamp rather than a vendor date — 100% of the sample has a declared, not assumed, before-open or after-close flag. Features are fundamental changes differenced year-on-year and standardised unexpected earnings, each stamped with the moment it became public; a single guard refuses any panel where a feature post-dates the trade. Filing text is acquired but was not yet complete for the full universe when this run was made, so the textual half of the hypothesis is untested here and is switched off explicitly rather than left to impute silently.

2 · SIX HELD-OUT YEARS

Year	Train n	Test n	IC	IC t	Pred. bp	Real. bp	Calib.	Sharpe
2019	4,916	1,356	-0.035	-0.83	174	-48	-0.23	-3.62
2020	6,271	1,408	0.052	1.42	127	212	1.38	0.35
2021	7,679	1,448	0.046	0.81	213	173	0.87	-0.66
2022	9,125	1,489	0.046	1.30	212	204	1.04	1.54
2023	10,616	1,535	-0.029	-0.98	229	-12	-0.09	-1.57
2024	12,147	1,565	0.055	2.21	179	-48	-0.21	-2.08

Table 1. Each row is a year the model never saw in training. *Pred.* and *Real.* are the predicted and realised top-minus-bottom quintile spreads in basis points; *Calib.* is the slope of realised on predicted, where 1.00 would mean the magnitudes were exactly right. The model predicted a positive spread of 127–229 bp every single year and realised between -48 and 212: confident throughout, and wrong as often as not. 2024 is the only year clearing $t = 2$ — one year in six, which is what a 5% threshold produces from noise — and it is also one of the worst for realised profit. Ranking and sizing are different skills.

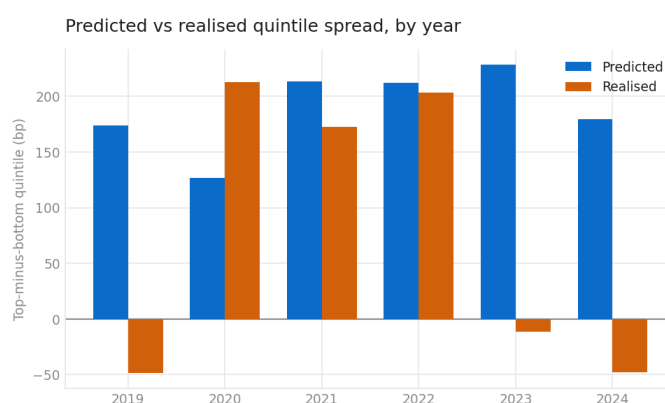


Figure 1. Predicted against realised top-minus-bottom quintile spread, by held-out year. A model with no signal still has opinions.

3 · WHY BELIEVE THE NULL

A null result and a broken pipeline look identical from outside, so the same code was run twice more on synthetic markets where the answer is known in advance. With a post-announcement drift planted in the data it is recovered in 6/6 years at a net Sharpe of 4.82. With nothing planted it returns 0.022 and 0.37 — yet still posts an IC of 0.080 at $t = 3.67$ in one individual year, a false positive in data where there is provably nothing to find. Anyone reporting a single year would have reported that one; the same caution applies to 2024 in Table 1.

Statistic	Real data	Synthetic null
Mean out-of-sample IC	0.022	0.022
t across the six years	1.30	1.15
Years with positive IC	4/6	4/6
Mean realised spread (bp)	80	89
Mean calibration slope	0.46	0.27
Net Sharpe, years stitched	-0.79	0.37
Newey–West t on daily P&L	-1.98	0.70

Table 2. The real study beside the synthetic null control. They look alike, which is the point: the real result is consistent with there being nothing to find.

4 · ALPHA, MULTIPLE TESTING, AND A SECOND METHOD

Regressed on Fama–French five factors plus momentum with Newey–West errors, the book shows an annualised alpha of -1.67% ($t = -2.16$) — no alpha. What the regression does find is significant loadings on profitability (RMW, $t = 4.08$), conservative investment (CMA, $t = 3.54$), value (HML, $t = 2.91$) and negative momentum (MOM, $t = -2.88$): the strategy is a quality-and-value portfolio in disguise, and those are compensated factors available for a few basis points. That alpha should also be read against the shuffled-prediction null in the abstract rather than against zero. Eight specifications are logged, including the two abandoned, giving a deflated Sharpe ratio of 0.03. Fama–MacBeth over 60 monthly cross-sections finds no feature significant at 5%, agreeing with the portfolio sort — which is the outcome that should raise confidence in a null rather than lower it.

5 · THE LIMITATION THAT MATTERS MOST

A point-in-time universe is not enough if the price source cannot serve delisted names. Yahoo returns no price history for 61% of the names deleted from the index during the sample, against 5% of the survivors, so 466 of the 735 members could be studied at all. The names lost include SIVB and FRC — Silicon Valley Bank and First Republic, both of which failed in 2023. Survivorship bias therefore re-enters through the data source even though the universe definition excludes it, and its direction is knowable: the missing firms are disproportionately those that collapsed, so the true result is probably somewhat worse than reported. Separately, SEC XBRL tags quarterly facts inconsistently. Differencing year-to-date flows into discrete quarters roughly doubled cash-flow coverage, but it still runs 40–50% rather than 100%, and the release corpus was incomplete at run time, leaving the textual half untested. Fixing the price source needs CRSP or an equivalent with delisting coverage.

6 · REPRODUCIBILITY

make reproduce regenerates every figure and number here and hashes each artefact; make verify diffs against the committed manifest, and CI fails if a hash moves. Method, the falsification criteria fixed before the run (none of the five is met), and a catalogue of the ways this study could flatter itself: github.com/pingadom/Earnings-Event-Research.